

CHEATSHEET · QUICK REFERENCE

Lab 14 — High Availability: Clustering, Load Balancing, NIC Teaming

Load-balance two backends with HAProxy, observe failover and failback, sketch keepalived VRRP, and compare NIC-teaming modes and redundancy layers.

HAProxy load balancing

```
balance roundrobin
Round-robin distribution

sed -i 's/balance roundrobin/balance leastconn/' /etc/haproxy/haproxy.cfg
Switch to least-connections

option httpchk GET /
Health-check the backends

systemctl restart haproxy
Apply config

for i in 1 2 3 4; do curl -s 127.0.0.1:8080; done
See alternation across backends
```

keepalived VRRP heartbeat

```
state MASTER / priority 150
Primary owns the VIP

virtual_ipaddress { 10.99.99.99/24 }
Floating VIP

advert_int 1
Heartbeat every 1 second
```

NIC teaming / bonding

```
modprobe bonding mode=active-backup miimon=100
Failover bond

ip link add bond0 type bond mode active-backup miimon 100
Create bond0
```

Redundancy layers

Layer	Component redundancy	Server-level redundancy
Power	Dual PSU	Two servers on separate UPS
Disk	RAID	Two servers with replication
Network	NIC teaming	Two servers behind LB + VIP
Compute	ECC RAM	N+1 cluster nodes